

Sprint 6 – Demo Readiness Checklist

This document verifies that your system is prepared for:

FINAL RELEASE READINESS VERIFICATION

Sprint 6 is not a normal Sprint Review.

This review evaluates whether your system is:

- stable
- repeatable
- integrated
- regression-safe
- presentation-ready

Incomplete or inaccurate checklists will impact evaluation.

Section 1 - Sprint Goal & Release Candidate Summary

Sprint Goal (Exact wording you will say during demo):

By the end of Sprint 6, a user will be able to repeatedly execute Momentum's full end-to-end flow (log in, browse and filter trails by difficulty and surface type, open and close a trail detail modal, log a ride to the activity calendar, and view and CO2 and money savings).

Release Candidate Summary

In 1–2 sentences, describe what a user can now reliably do in the product.

(Must reflect a stable, repeatable workflow – not planned functionality).

A user can log into Momentum, browse and filter the trail list by difficulty and surface type across consecutive filter changes without stale cards, open a trail detail modal and close it via clicking outside the modal, log a ride to the activity calendar, and view CO2 and money savings in the Calculator. This workflow completes reliably in repeated back-to-back sessions without refreshing a page or resetting the application.

Final Presentation Readiness Status

Ready for final presentation now

Explain:

The three named Sprint 6 bug fixes (the Trail Browsing Filter useEffect, the modal outside click closing mechanism, and the Calculator money savings bar scaling) have all been committed to main and have each passed our two-pass walkthrough rule for Sprint 6 on at least two separate class periods. The full end-to-end flow runs twice consecutively on a clean logged out browser without verbal explanation needed. Additionally, no experimental features are in the demo path.

Section 2 – Full Demo Workflow Plan

List the COMPLETE workflow that will be demonstrated.

This must match the actual live demo exactly.

The workflow should demonstrate:

- integration between systems
- realistic user interaction
- stable behavior
- repeatable execution

Add rows as needed.

Step Number	User Action	System Response
1	User opens Momentum and logs	Authentication succeeds; user lands

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	into the demo account.	on the home screen with nav bar visible.
2	User navigates to Browse Trails and applies the Easy difficulty filter, then switches to Moderate, then Hard, then back to Easy.	Trail card list re-renders cleanly on every filter switch with no stale cards from the prior selection
3	User taps a trail card to open the detail modal, then closes it by clicking outside the modal	Modal opens for the selected trail; outside-click closes the modal consistently. User repeats with two more trail cards in the same session
4	User navigates to Activity Log and logs a ride with today's date and a distance value	Ride entry is saved and appears in the calendar grid. Entry persists after navigating away and returning
5	User opens the Calculator, enters a distance, and views the CO2 and money savings output	CO2 chart and money savings bars render proportionally without requiring verbal explanation across all transit mode comparisons

Section 3 – Full Workflow Validation

These checks must be completed BEFORE submission

- ☒ Full workflow completed successfully from start → finish
- ☒ Workflow completed successfully at least TWICE consecutively
- ☒ Workflow tested using variation in user interaction or input
- ☒ Workflow tested without restarting application/system state
- ☒ Workflow tested using the actual demo device/environment
- ☒ All integrated systems remained functional during testing
- ☒ Previously working functionality was re-tested after recent changes
- ☒ Team verified no incomplete or experimental features are included in demo flow

Describe the MOST important workflow that was re-tested for regression prevention

The most important workflow that was re-tested for regression prevention was: Trail Browsing Filter (Easy -> Medium -> Hard -> Easy) & the Trail Detail Modal Open/Close; Activity Log Entry + Calculator CO2 and Money Savings. This is the exact sequence that failed on the Sprint 5 Demo and is the primary regression target for Sprint 6.

What specifically was verified during this regression check?

(Check all that apply)

- ☒ Data persistence
- ☒ Authentication/login
- ☒ Navigation flow
- ☒ API behavior
- ☒ Database interaction
- ☒ UI behavior
- ☒ State management
- ☐ Save/load behavior
- ☐ Multiplayer/shared state

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- ☐ Error handling
- ☐ Other: _____

Result of regression validation:

The result of regression validation was a pass. Consecutive filter switches now produce a clean re-render every time with no stale trail cards. The trail detail modal also closes consistently on outside-click across all trail cards in the same session. The Activity Log calendar grid is unaffected by the filter refactor. A full sequence was completed twice consecutively on a clean logged-out browser with no refresh or rollback.

Section 4 - Stability & Repeatability Verification

Sprint 6 evaluates whether your system behaves consistently under repeated normal use.

Stability Verification

- ☒ Key workflow repeated successfully multiple times
- ☒ No crashes, freezes, or resets occurred during testing
- ☒ No hidden setup was required between actions
- ☒ No “perfect path” conditions required
- ☒ System handled normal user interaction consistently
- ☒ System behavior remained predictable across repeated use
- ☒ Team verified the system can recover gracefully from small user mistakes

Highest Remaining Stability Risk

What is the SINGLE biggest remaining stability risk in your system?

The SINGLE biggest remaining stability in our system is the Trail Browsing filter logic under repeated use (TrailBrowse.tsx). Even after the useEffect refactor, rapid consecutive filter changes in a live demo could expose under edge-case state behavior not covered in the fixed dependency array. This is the exact failure mode from the Sprint 5 Demo.

What would happen during the demo if this failed?

If this failed during the demo, then stale trail cards from prior filter selections would remain visible after switching filters, exactly replicating the failure of our Sprint 5 Demo. The demo would then require a visible page refresh, which would signal instability and breaks the end-to-end flow.

Section 5 – Integration Readiness Verification

Sprint 6 evaluates whether systems still work together reliably.

Integration Checks Completed

- ☒ Frontend ↔ backend integration tested
- ☒ Database/storage integration tested
- ☒ API/service integration tested
- ☒ Authentication/user state tested
- ☒ Cross-page/system navigation tested
- ☒ Shared state/data consistency verified
- ☒ Deployment/build process verified
- ☒ No known integration-breaking merge conflicts remain
- ☒ Team verified all major systems still function together after recent changes

Which integration is currently MOST fragile?

The integration that is currently MOST fragile is the Trail Browsing Filter State to the Activity Log (shared global state). The filter useEffect refactor in Task 1 touches a state slice both features depend on. A change to the filter dependency array could trigger unexpected re-renders in the Activity Log calendar grid, breaking a previously stable feature mid-demo.

What was done to reduce this risk before demo day?

The filter useEffect dependency array was changed from the derived filteredTrails slice to the raw difficulty and surfaceType state values. A pre-merge regression check was run on the Activity Log calendar grid (the known shared-state neighbor) before merging to main. The smoke test script documents both the filter sequence and the Activity Log verification step so any teammate can re-run the regression check independently.

Section 6 – Demo Repeatability Proof

You must visibly prove repeatability during the live demo.

Choose at least ONE visible proof of stability:

- ☒ Repeat a key workflow live
- ☒ Show variation in user behavior/input
- ☐ Demonstrate persistence across actions
- ☐ Demonstrate stable behavior after multiple operations
- ☐ Demonstrate integration between multiple systems live

Describe EXACTLY what I should see during this repeatability proof

(Be specific and observable)

Sathkrith will open a clean incognito browser, log into Momentum using our demo account, browse trails, apply the Easy filter, switch to Moderate, switch to Hard, then back to Easy. No stale cards will be visible at any step. Sath will open a trail detail modal and close it by clicking outside. Sath will then log a ride to the activity calendar. Sath will then open the Calculator and enter a distance. The CO2 and money savings bars should render without verbal explanation.

Section 7 – Demo Roles & Operational Readiness

Demo Lead: Sathkrith

Secondary Support: Yohaana

Readiness Confirmation

- ☒ Demo Lead has personally completed the full workflow before class
- ☒ Secondary Support understands all demonstrated systems and integrations
- ☒ All team members understand what is stable, what is fragile, and what release risks still remain
- ☒ Team can answer questions about stability, integration, testing, and regression prevention
- ☒ Team understands that explanation does NOT replace working functionality

Section 8 – Demo Environment & Release Branch Verification

Environment Readiness

- ☒ Devices ready before demo begins
- ☒ Application launches immediately

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- ☒ Required accounts/logins prepared
- ☒ Demo environment tested before class
- ☒ Internet/network dependencies verified
- ☒ No setup steps required during demo
- ☒ Demo system matches release branch state

Sprint 6 Release Branch Verification

- ☒ Release branch represents EXACT demo version
- ☒ No experimental or incomplete code included
- ☒ No additional development planned before demo
- ☒ Team understands release branch is a time-locked release snapshot

Section 9 – Professional Demo Standards

During the demo and Q&A you may NOT display:

- planning documents
 - notes/scripts
 - IDE debugging output
 - non-product materials
 - hidden setup steps
- ☒ We understand that only the product and required product interaction should appear during the demo
- ☒ We understand that explanation does NOT replace missing functionality
- ☒ We understand that unstable or incomplete systems are not considered release-ready
- ☒ We understand that demo-time debugging or fixes are not allowed

Section 10 - Final Release Readiness Certification

Each team member is accountable for the accuracy of this document.

By checking these boxes, your team confirms:

- ☒ This system can be demonstrated start → finish without resets, retries, or explanation
- ☒ This system behaves consistently under repeated use
- ☒ Previously working functionality has been re-tested after recent changes
- ☒ Integrated systems function together reliably
- ☒ This product is capable of supporting a professional final presentation
- ☒ This demo represents the true release state of the product
- ☒ This checklist accurately reflects what will occur during the live demo